

NMAP (NETWORK MAPPER) CHEAT SHEET

Essential Network Discovery & Vulnerability Assessment Commands for Security Students

1. Host Discovery (Finding Active Devices)

Always identify live systems before initiating deep-port scanning to save time and reduce target network noise.

Command Syntax	Core Action	Why & When to Use It
<code>nmap -sn [target]</code>	Ping Sweep	Discovers live hosts across a subnet without scanning individual ports. Fast asset inventory.
<code>nmap -Pn [target]</code>	Skip Ping Probe	Forces Nmap to scan all ports directly, treating the host as online. Essential when firewalls block ICMP.

2. Scanning Techniques (Target Connection Types)

These flags alter the handshake behavior between Nmap and target socket interfaces.

Command Syntax	Scan Type	How It Works & Student Notes
<code>sudo nmap -sS [target]</code>	SYN Scan (Stealth)	Default/Best choice. Sends a SYN packet; cuts off connection with RST once SYN-ACK returns. Bypasses old loggers. (Requires Root).
<code>nmap -sT [target]</code>	Connect Scan	Completes full TCP 3-way handshake (SYN → SYN-ACK → ACK). Slower, fully logged, but works without root privileges.
<code>sudo nmap -sU [target]</code>	UDP Scan	Probes connectionless UDP ports (DNS: 53, SNMP: 161). Slow but critical; waits for timeout or ICMP unreachable.

3. Port Selection & Range Filtering

Narrow down target ranges to expedite audits, or expand them to ensure zero security blindspots.

Command Syntax	Target Scope	Why & When to Use It
<code>nmap -F [target]</code>	Fast Scan (Top 100)	Scans only the top 100 most commonly used ports. Super fast initial triage.
<code>nmap -p [ports] [target]</code>	Specific Ports	Specify list or ranges: e.g., <code>-p 22,80,443,8080</code> or range <code>-p 1-1024</code> . Saves scan time.
<code>nmap -p- [target]</code>	All 65,535 Ports	Scans every single port available. Crucial for detecting backdoors or custom non-standard service ports.

4. Service Fingerprinting & Deep-Dive Recon

Go beyond simple open/closed status. Identify the exact software, version, and OS running behind active ports.

Command Syntax	Audit Objective	Key Advantages
<code>nmap -sV [target]</code>	Service Versioning	Banner grabs and probes sockets to pinpoint software versions (e.g., Apache 2.4.49). Essential for checking CVEs.
<code>sudo nmap -O [target]</code>	OS Detection	Analyzes network protocol responses (TCP/IP stack behavior) against database to guess operating systems.
<code>nmap -A [target]</code>	Aggressive Scan	A comprehensive shorthand switch combining: OS detection (<code>-O</code>), Service Version (<code>-sV</code>), Default scripts (<code>-sC</code>), and Traceroute.

5. Nmap Scripting Engine (NSE)

Automate specialized auditing tasks using built-in, highly targeted Lua scripts.

Command Syntax	Category/Purpose	Use-Case Scenario
<code>nmap -sC [target]</code>	Default Script Scan	Runs a safe suite of lightweight scripts (equivalent to <code>--script=default</code>).
<code>nmap --script safe [target]</code>	Safe Suite	Executes passive, non-intrusive scripts unlikely to crash old or fragile targets.
<code>nmap --script vuln [target]</code>	Vulnerability Check	Checks identified services against databases of known flaws (Shellshock, Heartbleed, etc.).
<code>nmap -p 443 --script ssl-cert [target]</code>	Targeted Script	Extracts and inspects the SSL certificate on port 443 (e.g. valid dates, subdomains).

6. Timing & Speed Management

Manage probe rates using timing templates to balance speed against detection evasion.

T0 - T1

Sneaky & Slow

Extremely slow scans (T0: paranoid, T1: sneaky). Use to evade strict firewall systems and Intrusion Detection Systems (IDS).

T4 - T5

Fast & Insane

T4 (aggressive) is highly optimized for CTFs, labs, and modern fast networks. T5 (insane) sacrifices packet accuracy for maximum speed.

T2 - T3

Polite & Default

T2 (polite) throttles bandwidth to respect system health. T3 (normal) is the default rate used if nothing is specified.

7. Evasion, Output, & Student Strategy

 Student Golden Rule (The 3-Step Scan Workflow):

- 1. **Subnet Sweep:** Find live hosts fast: `nmap -sn 192.168.1.0/24`
- 2. **Fast Port Find:** Identify active ports on a host: `nmap -T4 -F 192.168.1.10`
- 3. **Full Inspection:** Audit everything, output findings: `nmap -T4 -p- -A -oA scan_report 192.168.1.10`

Useful Parameter	Evasion / Utility	What It Accomplishes
<code>sudo nmap -f [target]</code>	Packet Fragmentation	Splits TCP headers into tiny 8-byte fragments to slip past basic Firewalls.
<code>sudo nmap -D RND:10 [target]</code>	Decoy IP Spoofing	Generates 10 decoy traffic sources alongside your real IP to confuse SOC analysts.
<code>nmap -oA [filename] [target]</code>	Full Output Logging	Saves findings in three formats: standard text (<code>.nmap</code>), Grepable (<code>.gnmap</code>), and XML (<code>.xml</code>).